

EXPERIMENT 2

Comparative Study of Zoho Creator and Traditional Software

1. Aim

To study and compare the development of applications using Zoho Creator with the traditional software development approach, focusing on coding, development time, customization, maintenance, and deployment.

2. Introduction

Zoho Creator is a low-code application development platform that provides visual tools for building forms, workflows, reports, and applications. Traditional software development generally involves writing and maintaining application code using programming languages, frameworks, databases, and development tools. This experiment compares the two approaches to understand their major differences.

3. Objectives of the Study

- To understand the basic features of the Zoho Creator development approach.
- To study the major characteristics of traditional software development.
- To compare coding requirements and development effort.
- To compare development time, customization, maintenance, and deployment.
- To identify the practical advantages and limitations of both approaches.

4. Procedure

1. Study the main features and working approach of Zoho Creator.
2. Observe how forms and application components can be created using its visual builder.
3. Review the traditional approach involving source code, application logic, and database configuration.
4. Compare both approaches based on development time, coding effort, customization, maintenance, and deployment.
5. Record the observations in a comparative table.
6. Identify suitable situations in which each approach can be applied.

5. Comparative Study

The two approaches differ mainly in the amount of programming required and the way application components are developed and maintained. Zoho Creator provides a visual, low-code environment, whereas traditional development generally requires direct programming and configuration of software components.

5.1 Zoho Creator Approach

- Uses visual form and application builders.
- Requires comparatively less programming for basic applications.
- Allows fields, workflows, and reports to be configured through the platform.
- Supports faster development and easier prototyping for suitable applications.
- Deployment and access can be handled through the platform.

5.2 Traditional Software Development

- Requires programming using languages and frameworks.
- Application logic, interfaces, and database components are generally developed separately.
- Development usually involves coding, testing, debugging, and configuration.
- Provides greater control over application architecture and customization.
- Deployment and maintenance require management of the developed software environment.

5.3 Comparison Table

Feature	Zoho Creator	Traditional Software
Development	Low-code / visual	Code-based development
Coding Requirement	Minimal for basic apps	Usually extensive
Development Time	Generally shorter	Generally longer
Customization	Platform-based options	High control through code
Maintenance	Simpler for basic apps	Requires development effort
Deployment	Platform-supported	Requires software environment setup
Database	Integrated platform support	Configured and managed separately as required
Scalability	Depends on platform capabilities	Depends on system architecture and implementation

5.4 Zoho Creator Example

A student information application can be created in Zoho Creator by arranging fields in the visual Form Builder. The screenshot below illustrates the kind of form-building interface used to configure fields and properties without developing every component manually.

5.5 Form Builder View

The screenshot displays the Zoho Creator Form Builder interface for a 'Hostel Complaint' form. The interface is divided into three main sections: a left sidebar for field selection, a central canvas for field arrangement, and a right sidebar for field properties.

Left Sidebar (Basic Fields): Contains icons for various field types: Name, Email, Address, Phone, Single Line, Multi Line, Number, Date, Time, Drop Down, Radio, Checkbox, Lookup, and File Upload.

Central Canvas: Shows the form layout with the following fields: 'Student Name' (split into First Name and Last Name), '123 Register Number', '123 Room Number', 'Hostel Block' (dropdown), 'Complaint Type' (dropdown), 'Complaint Description' (text area), 'Complaint Date' (calendar icon, format dd-MMM-yyyy), 'Priority' (radio buttons: Low, Medium, High, Urgent), 'Contact Number', and 'Complaint Status' (dropdown).

Right Sidebar (Field Properties): Allows configuration of the selected field ('Student Name'). It includes sections for: 'Field name' (Student Name), 'Field link name' (Student_Name), 'Hide field information' (link), 'Validation' (Mandatory checkbox), 'Display Fields' (Prefix, First Name, Last Name, Suffix checkboxes), and 'Data Privacy' (Contains personal data checkbox).

Bottom Navigation Bar: Includes tabs for Form, Reports, Pages, Workflow, and Settings, along with a Help icon and a Next button.

Figure 1: Zoho Creator Form Builder interface.

6. Advantages and Limitations

Zoho Creator – Advantages

- Visual and low-code development process.
- Faster creation of many basic applications.
- Built-in components reduce repetitive development work.
- Useful for rapid application development and prototyping.

Traditional Software – Advantages

- Greater control over system architecture and implementation.
- Extensive customization can be achieved through programming.
- Suitable for applications requiring specialized or complex functionality.

Limitations

- Zoho Creator customization is dependent on platform capabilities.
- Traditional development generally requires more coding, testing, and maintenance effort.

7. Result

Thus, the comparative study showed that Zoho Creator provides a visual, low-code approach for building applications, while traditional software development offers direct programming control and wider customization. The choice of approach depends on the application's requirements, complexity, and development constraints.